

Day 06 of 60

3 BUCKETS.

Everything ML falls into one of them.

Before learning specific algorithms, you need to know which bucket you're operating in.

 3 Core Types

 4 Min Read

 Interview Cheat Sheet

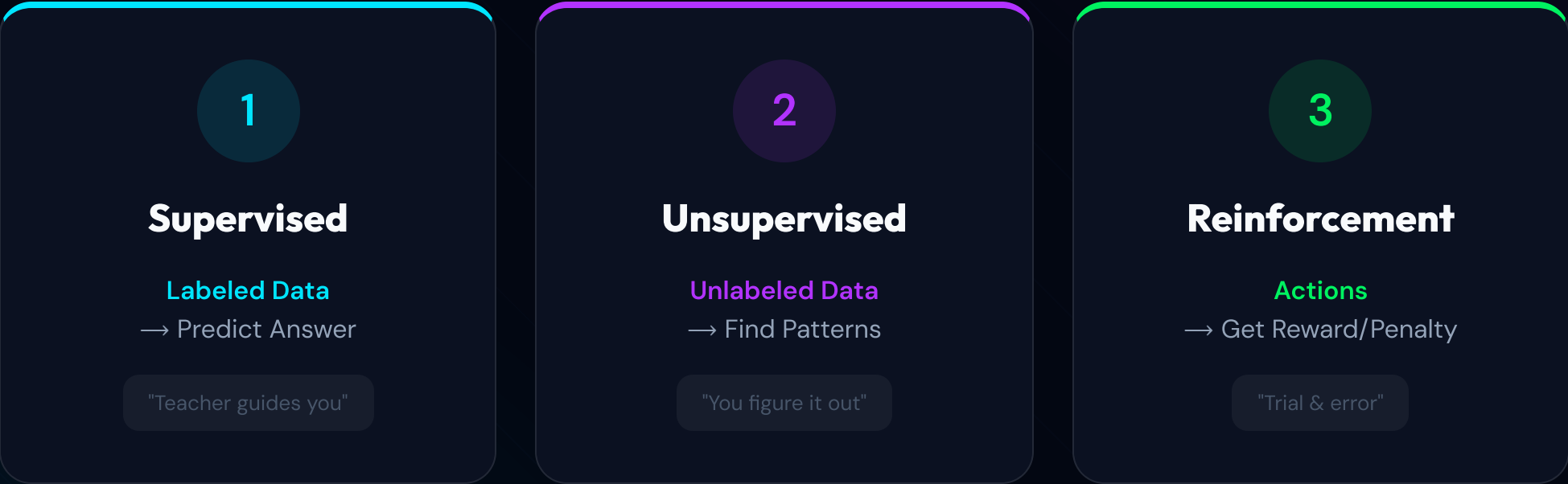


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The Visual Bucket Diagram



Supervised Learning

You have the answers (labels). You teach the model to map inputs to those answers so it can predict future unseen inputs. This is **90% of enterprise ML**.

Classic Examples:



House Price Prediction (Regression)

Input: Bedrooms, Square footage → Output: Price (\$)



Spam Detection (Classification)

Input: Email text → Output: "Spam" or "Not Spam"



Unsupervised Learning

You just have data. No answers. The model's job is to find hidden structure, groups, or anomalies inside the data entirely on its own.

Classic Examples:



Customer Segmentation (Clustering)

Input: Purchase history → Output: Grouped into distinct buyer personas.



Fraud Detection (Anomaly Detection)

Input: Millions of transactions → Output: Flag the 0.01% that look weird.



Reinforcement Learning (RL)

Learning by trial and error. An "agent" takes actions in an environment to maximize a "reward". It learns the best strategy over time.

Classic Examples:



Game Playing Agents

AlphaGo beating the world champion at the game of Go.



Self-Driving Cars & Robotics

Reward: Staying on the road. Penalty: Crashing.



Interview Cheat Sheet

TYPE	DATA USED	THE GOAL	SIMPLE ANALOGY
Supervised	Labeled	Predict outcomes	"Is this a picture of a cat or a dog?"
Unsupervised	Unlabeled	Find hidden patterns	"Group these 1000 users by behavior."
Reinforcement	No Data (Actions)	Maximize reward	"Figure out how to beat this video game."





✓ Day 06 Complete

Tomorrow on Day 07:

The Complete ML Workflow

From raw data collection to deploying a model to production.

Follow for the full 60-day ML series

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#SupervisedLearning

#AIConcepts

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